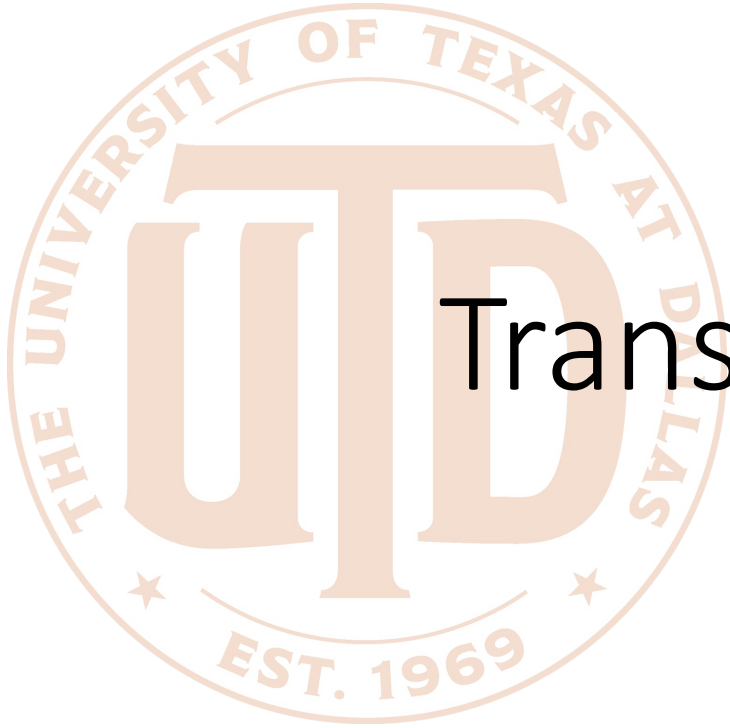




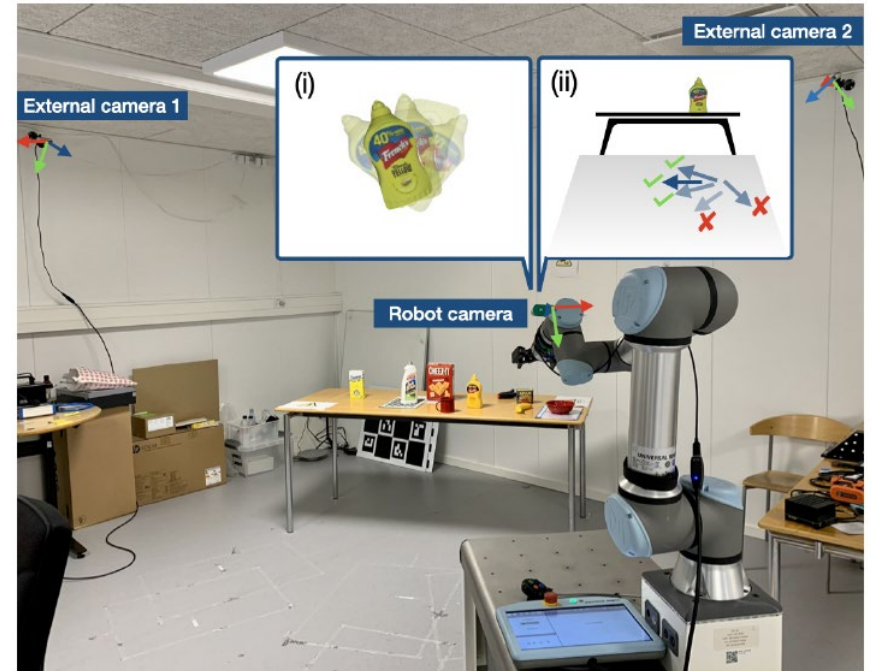
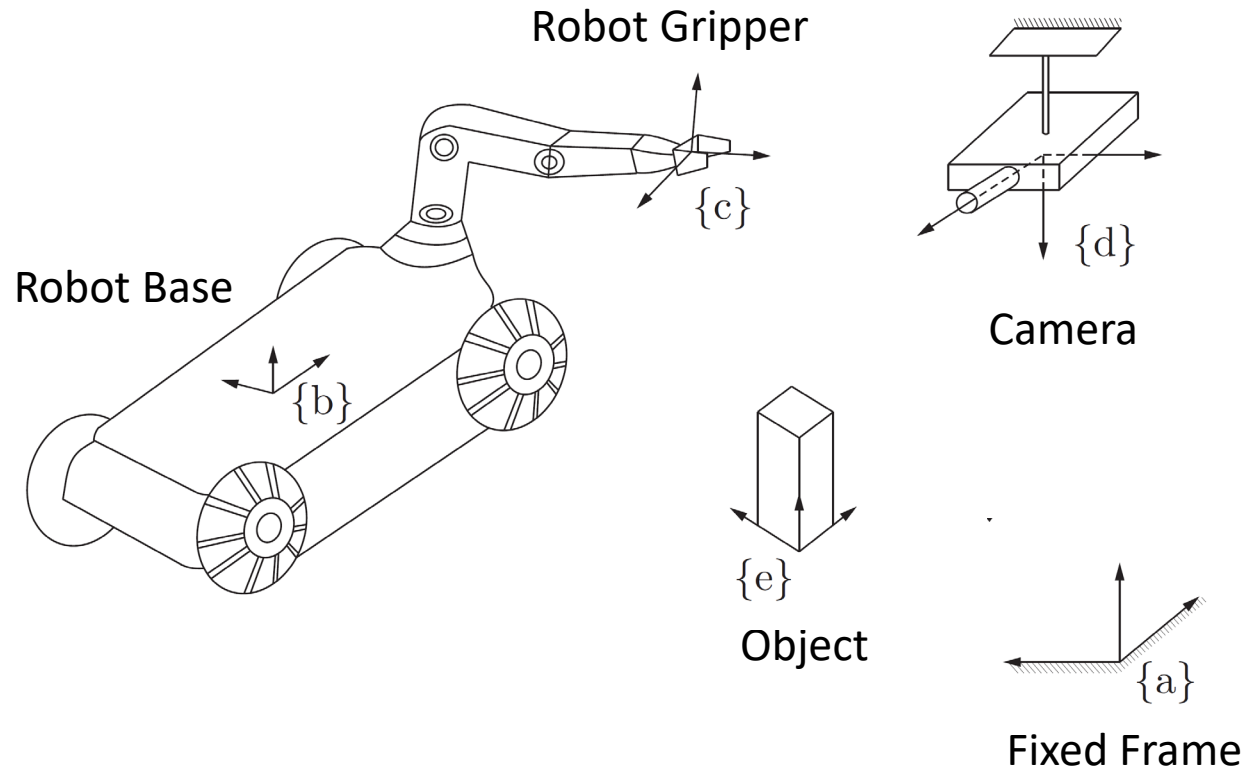
# Transformations and Forward Kinematics

CS 6341: Robotics

Instructor Yu Xiang

The University of Texas at Dallas

# Transformation Matrices in Robotics

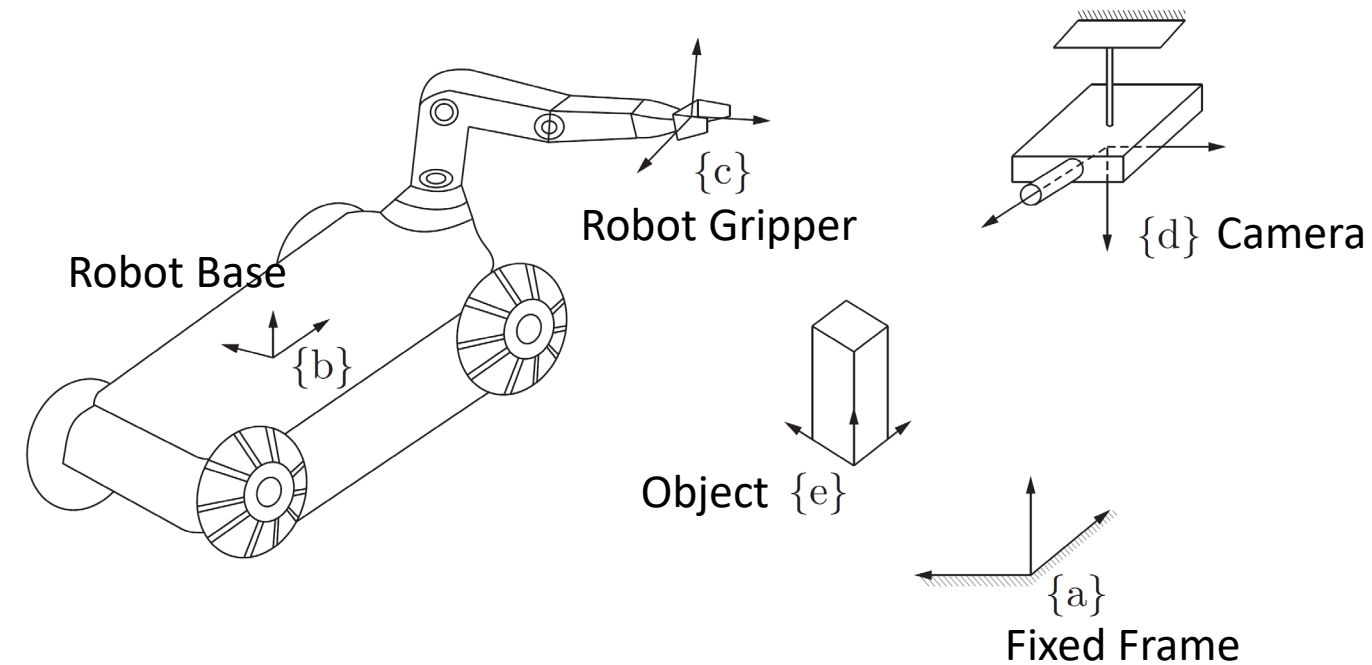


Multi-view object pose distribution tracking for pre-grasp planning on mobile robots. Naik et al., ICRA, 2022.

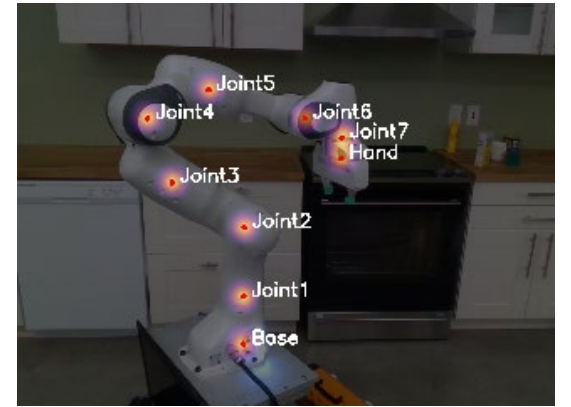
- How to move the robot arm to pick up the object?
- What information we need to know?

$$T_{ce} \quad \text{Pose of object in gripper}$$

# Transformation Matrices



- Robot tracking in camera

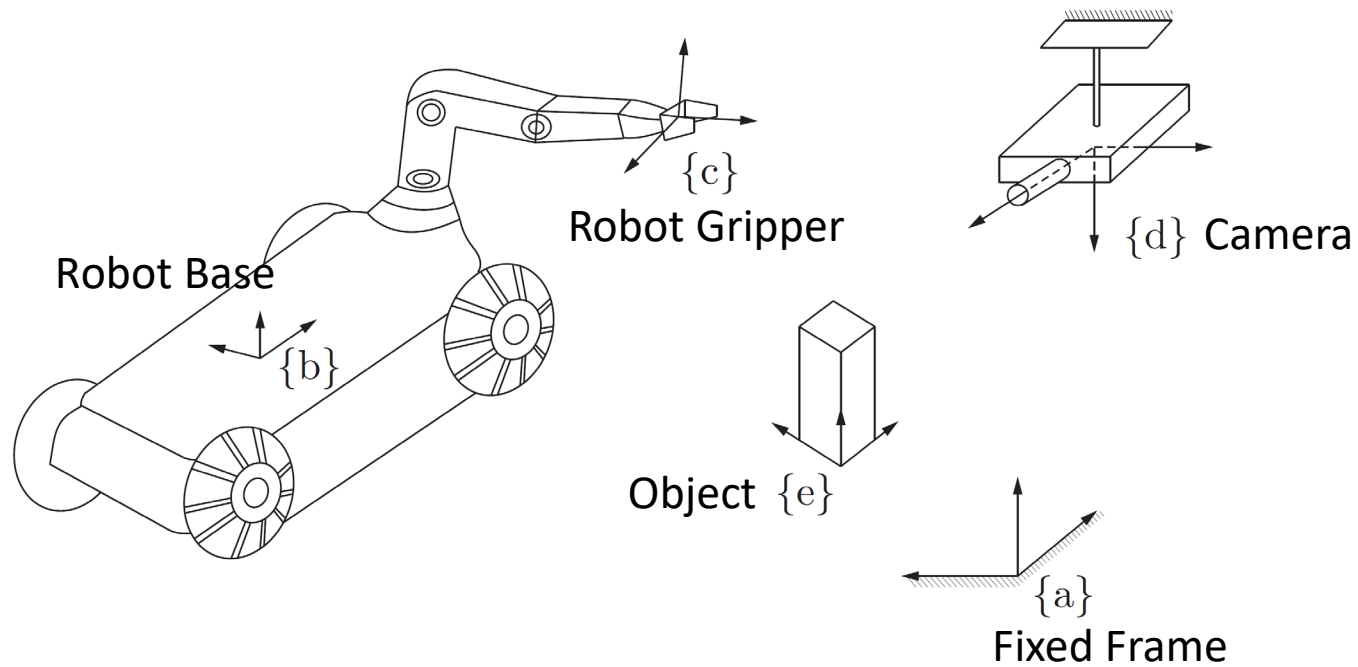


[https://research.nvidia.com/publication/2020-05\\_camera-robot-pose-estimation-single-image](https://research.nvidia.com/publication/2020-05_camera-robot-pose-estimation-single-image)

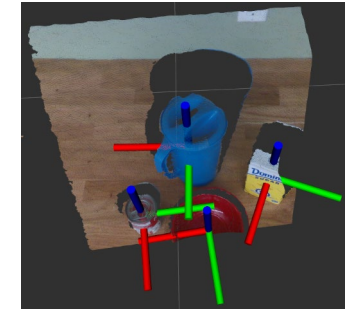
Robot in camera

$$T_{db} = \begin{bmatrix} 0 & 0 & -1 & 250 \\ 0 & -1 & 0 & -150 \\ -1 & 0 & 0 & 200 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

# Transformation Matrices



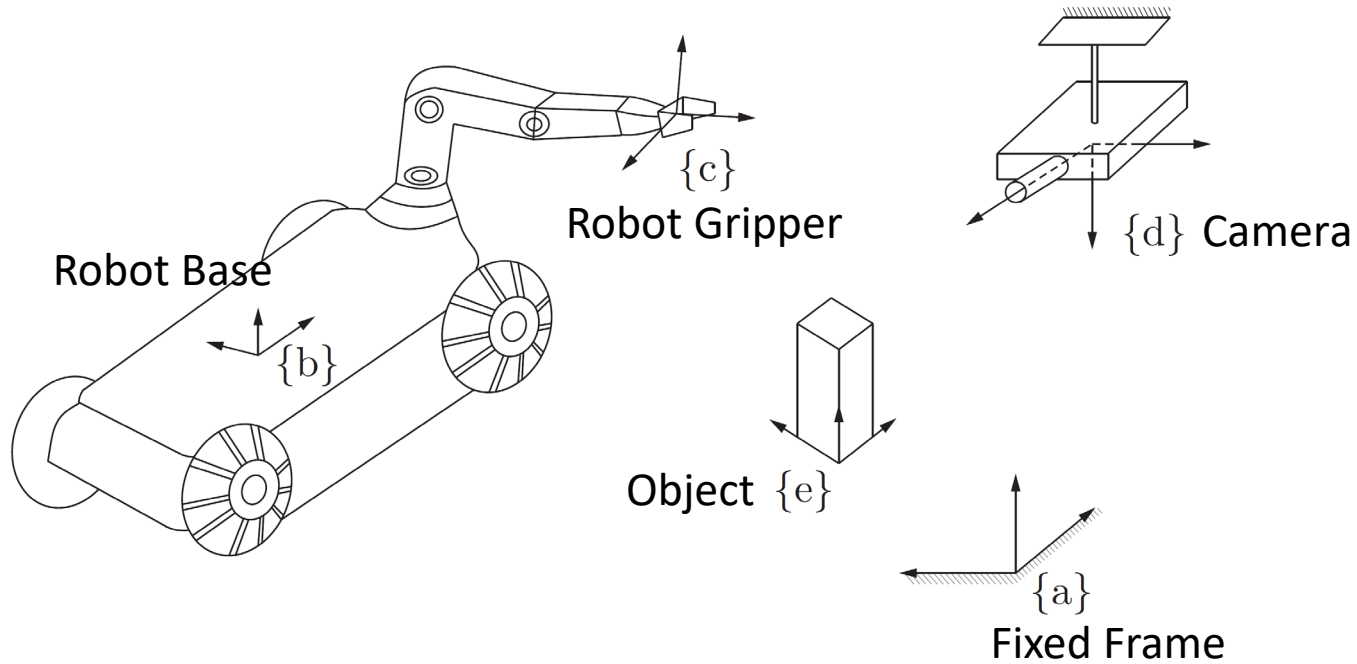
- Object pose estimation from camera



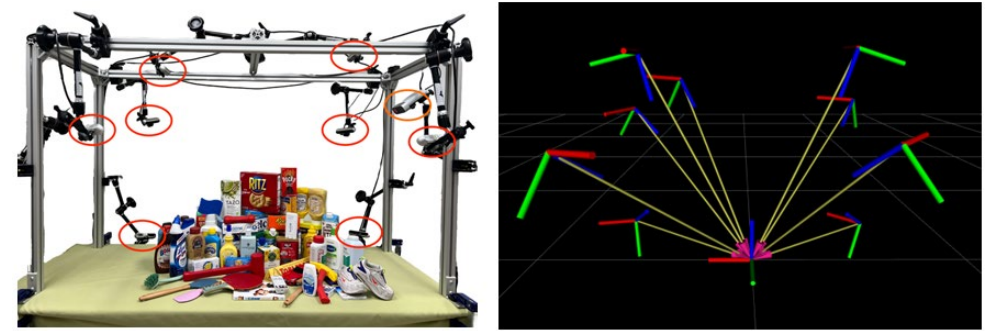
Object in camera

$$T_{de} = \begin{bmatrix} 0 & 0 & -1 & 300 \\ 0 & -1 & 0 & 100 \\ -1 & 0 & 0 & 120 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

# Transformation Matrices



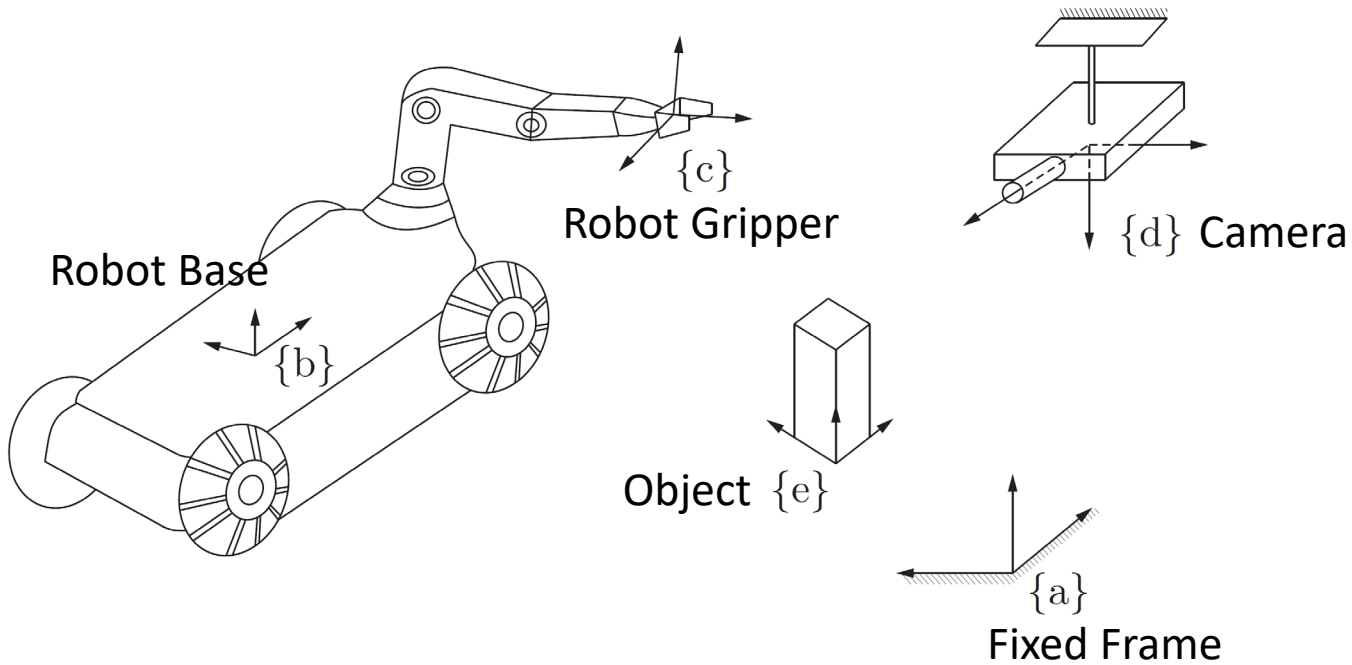
- Camera calibration



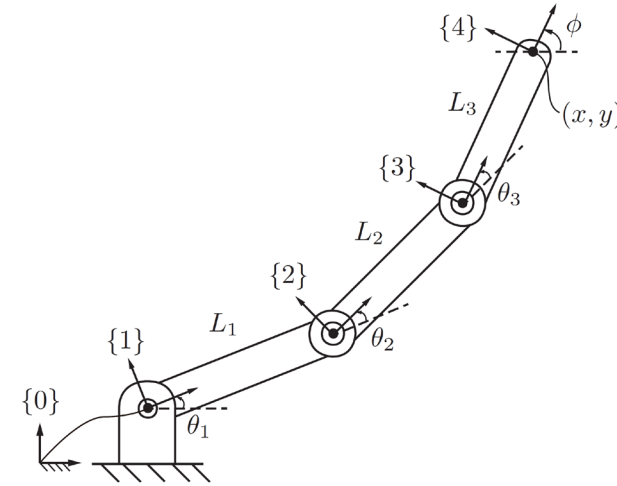
Camera in fixed frame

$$T_{ad} = \begin{bmatrix} 0 & 0 & -1 & 400 \\ 0 & -1 & 0 & 50 \\ -1 & 0 & 0 & 300 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

# Transformation Matrices



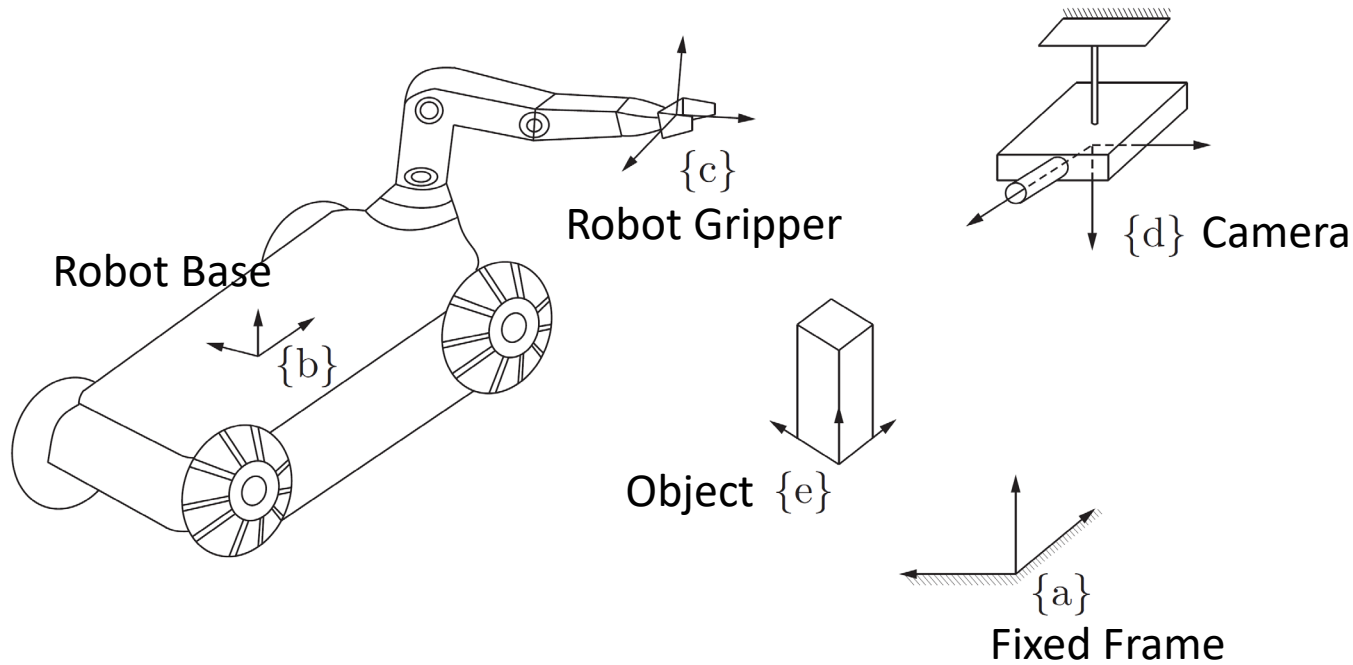
- Forward kinematics (future lectures)



Gripper in robot

$$T_{bc} = \begin{bmatrix} 0 & -1/\sqrt{2} & -1/\sqrt{2} & 30 \\ 0 & 1/\sqrt{2} & -1/\sqrt{2} & -40 \\ 1 & 0 & 0 & 25 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

# Transformation Matrices



- We know the following transformations

Robot in camera

$$T_{db} = \begin{bmatrix} 0 & 0 & -1 & 250 \\ 0 & -1 & 0 & -150 \\ -1 & 0 & 0 & 200 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Object in camera

$$T_{de} = \begin{bmatrix} 0 & 0 & -1 & 300 \\ 0 & -1 & 0 & 100 \\ -1 & 0 & 0 & 120 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

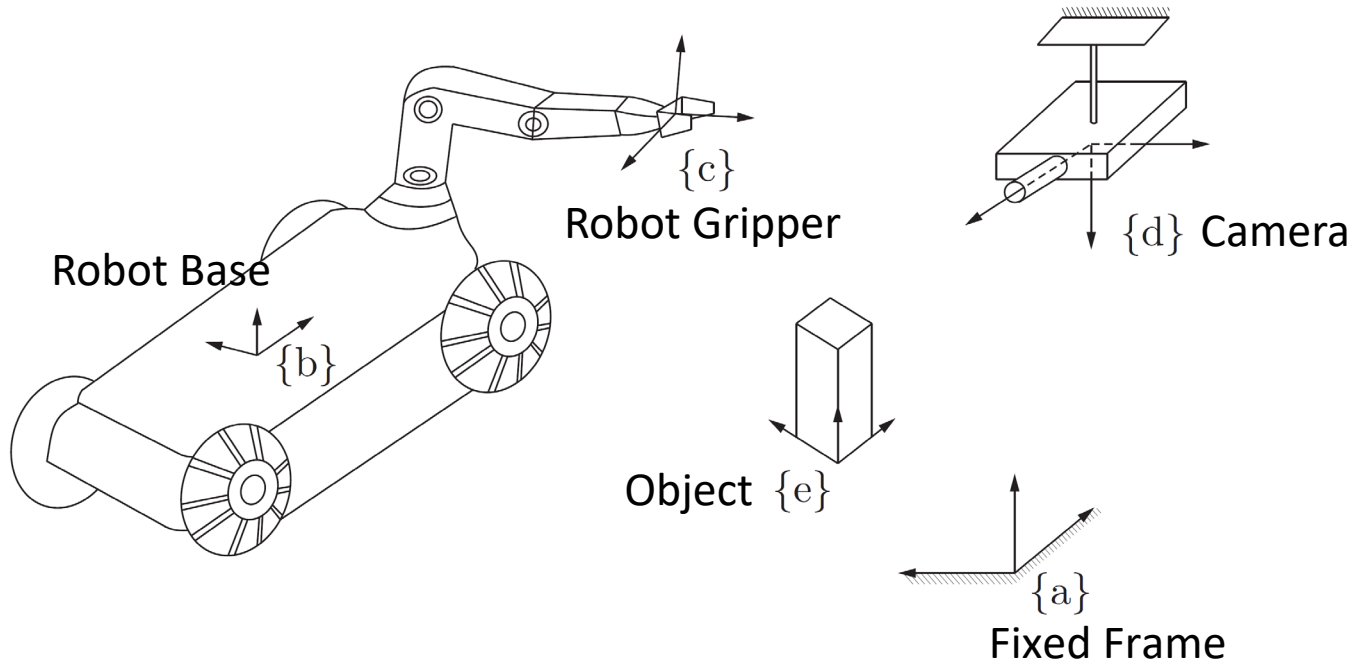
Camera in fixed frame

$$T_{ad} = \begin{bmatrix} 0 & 0 & -1 & 400 \\ 0 & -1 & 0 & 50 \\ -1 & 0 & 0 & 300 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Gripper in robot

$$T_{bc} = \begin{bmatrix} 0 & -1/\sqrt{2} & -1/\sqrt{2} & 30 \\ 0 & 1/\sqrt{2} & -1/\sqrt{2} & -40 \\ 1 & 0 & 0 & 25 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

# Transformation Matrices



- How to move the robot arm to pick up the object?

$$T_{ce}$$

- We know  $T_{db}$   $T_{de}$   $T_{bc}$   $T_{ad}$

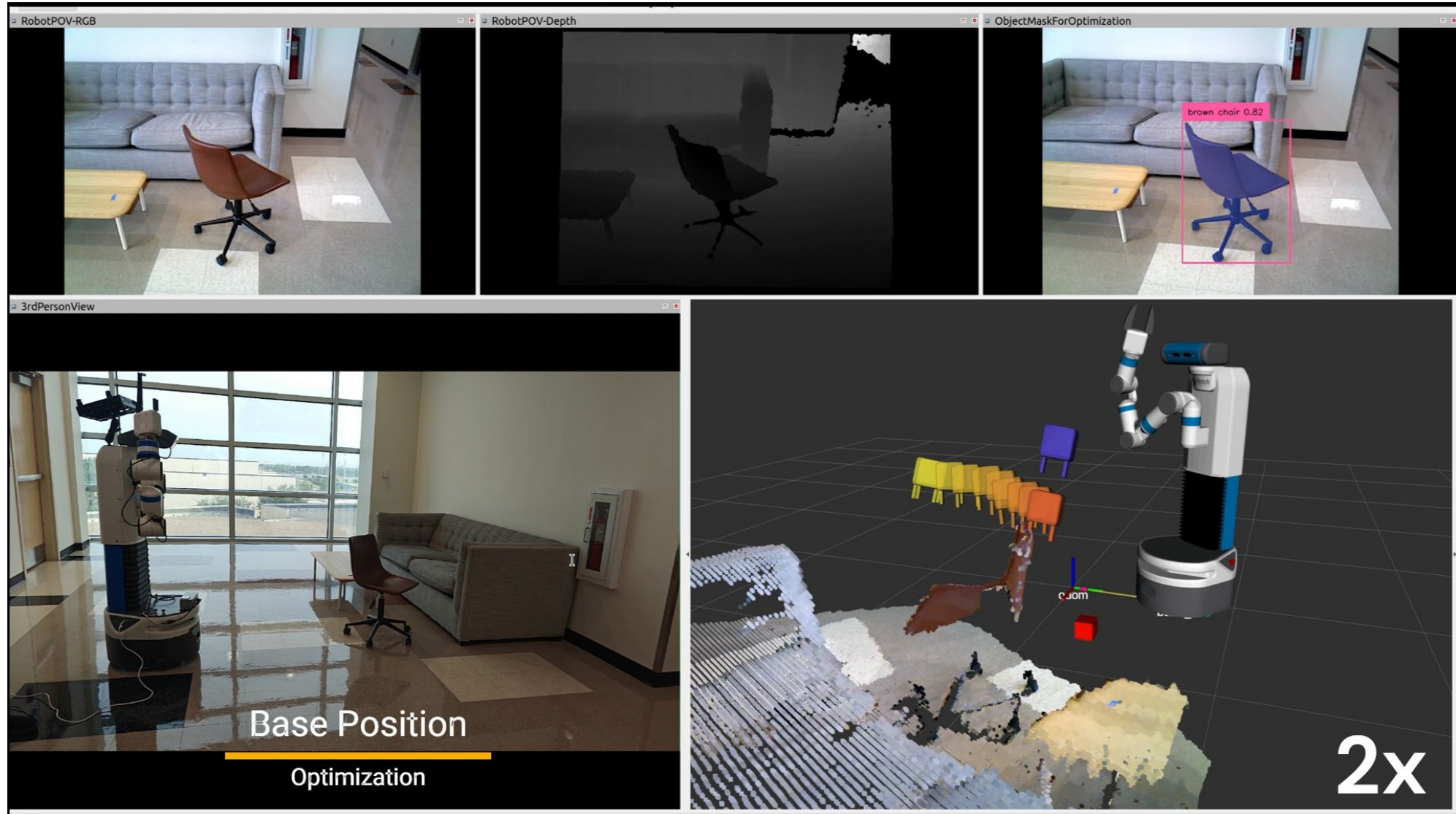
$$T_{ce} = \begin{bmatrix} 0 & 0 & 1 & -75 \\ -1/\sqrt{2} & 1/\sqrt{2} & 0 & -260/\sqrt{2} \\ -1/\sqrt{2} & -1/\sqrt{2} & 0 & 130/\sqrt{2} \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$T_{ce} = T_{bc}^{-1} T_{db}^{-1} T_{de}$$

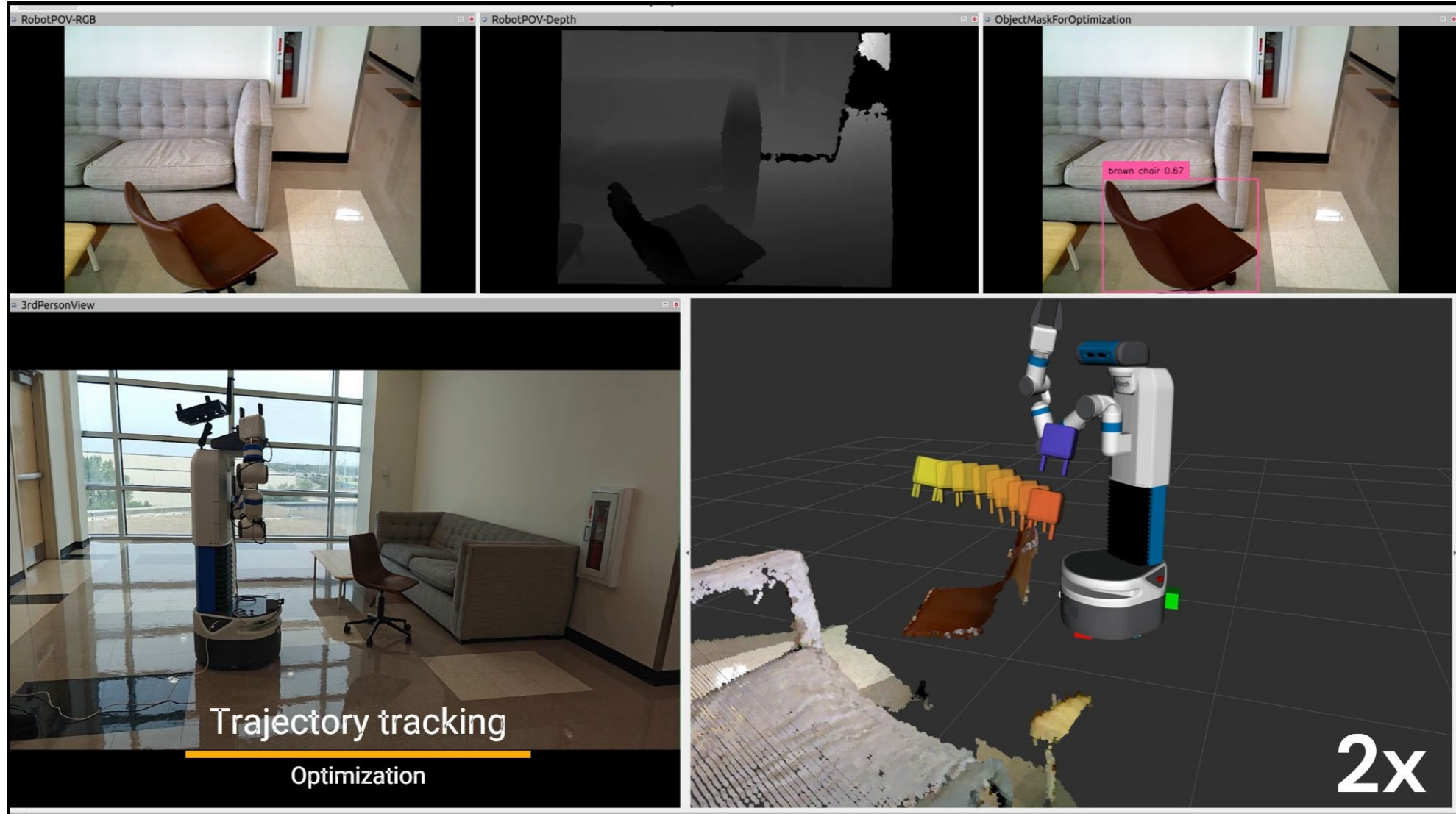
# Demo



# Optimizing the Robot Base Location

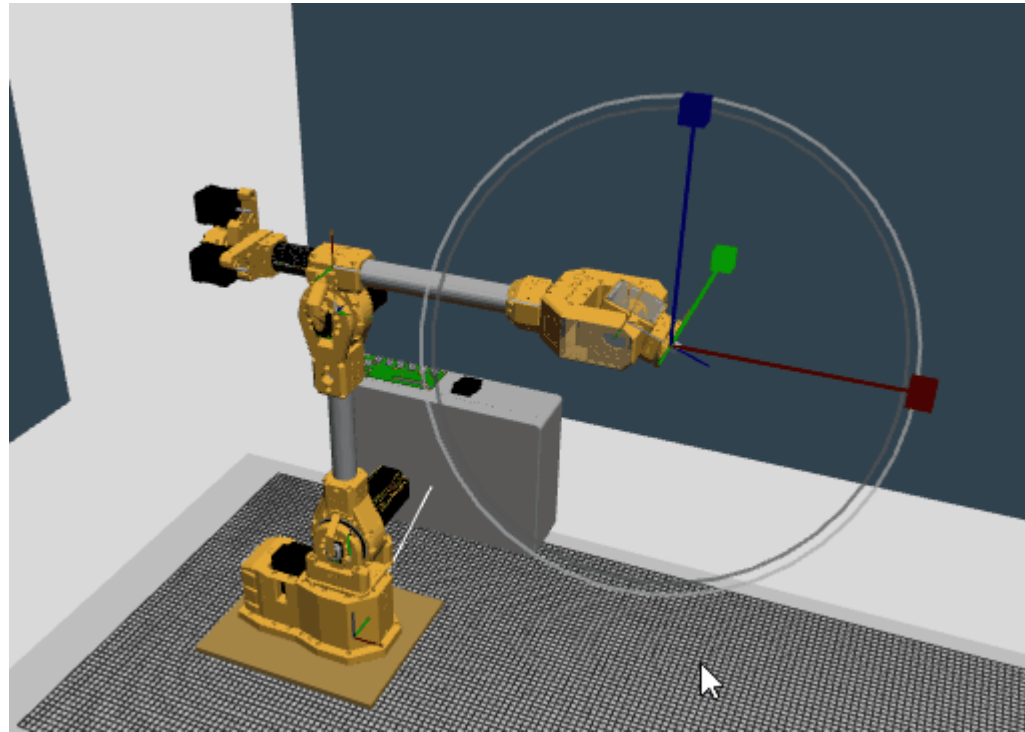


# Optimizing the Robot Trajectory



# Robot Kinematics

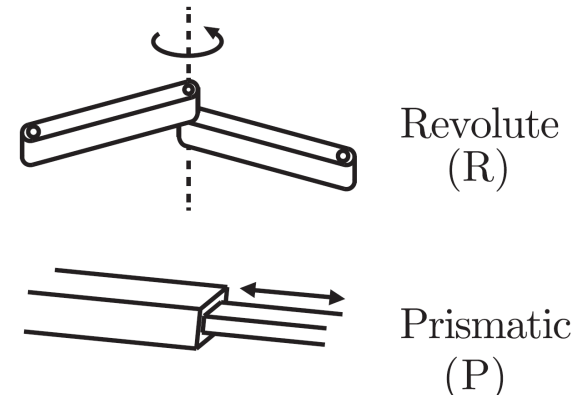
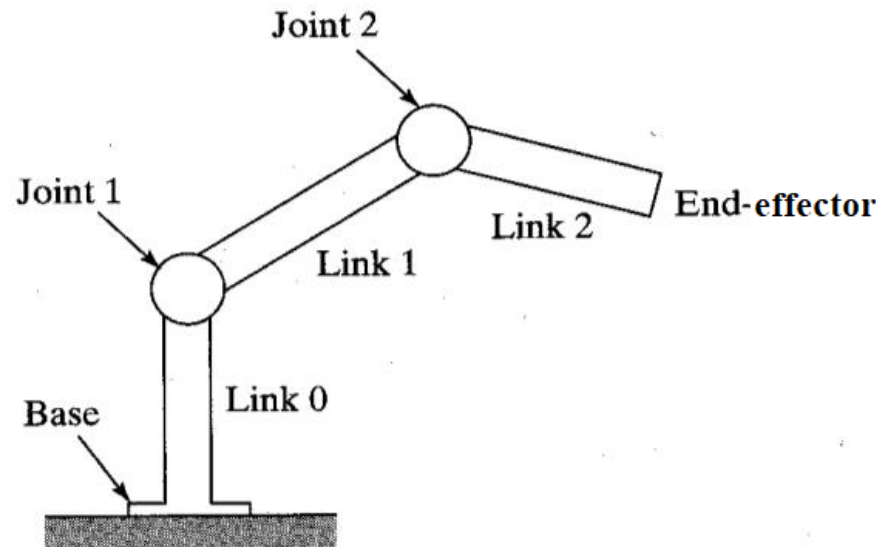
- The relationship between a robot's joint coordinates and its spatial layout



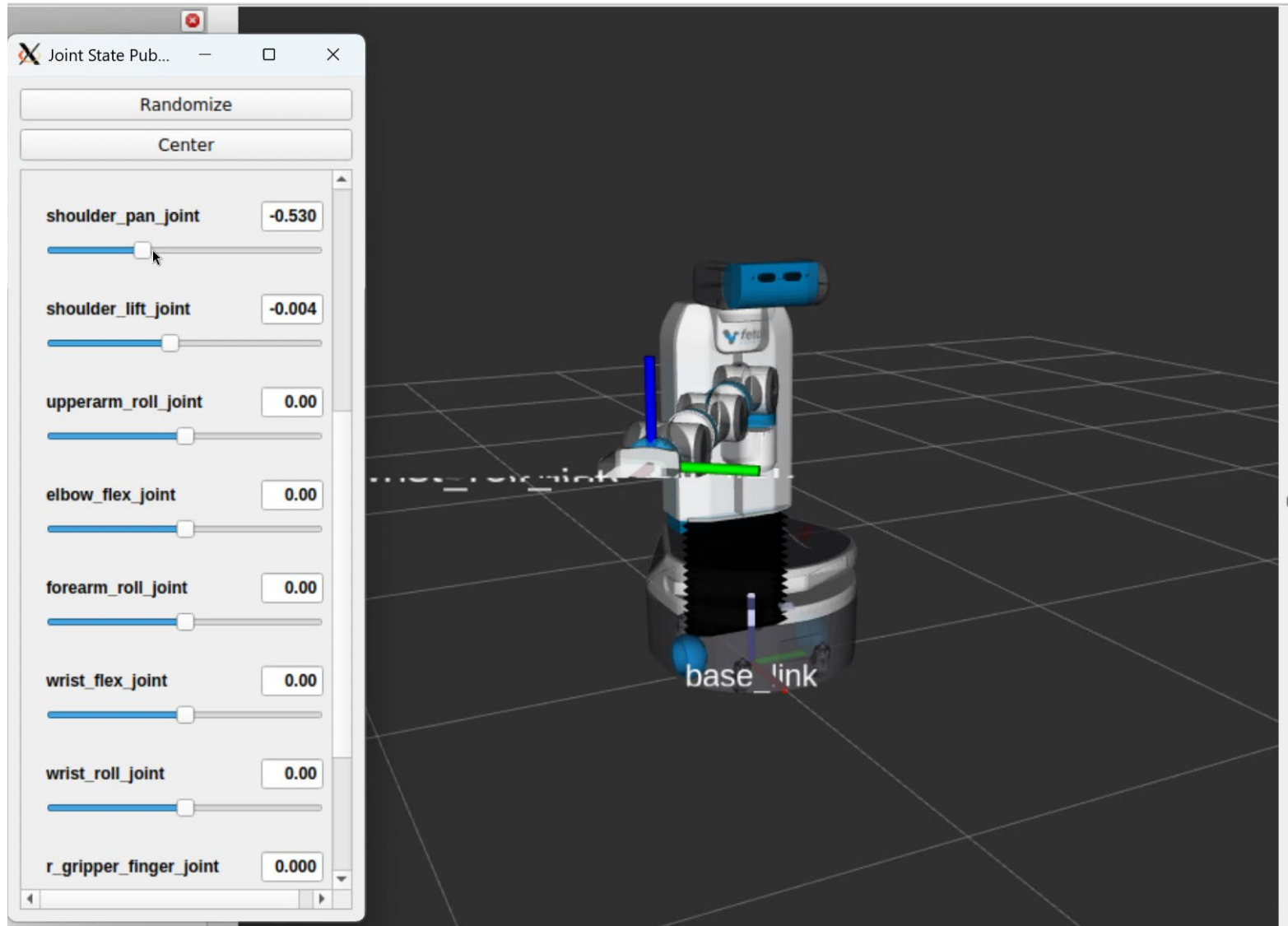
<https://www.marginallyclever.com/2020/04/gradient-descent-inverse-kinematics-for-6dof-robot-arms/>

# Forward Kinematics

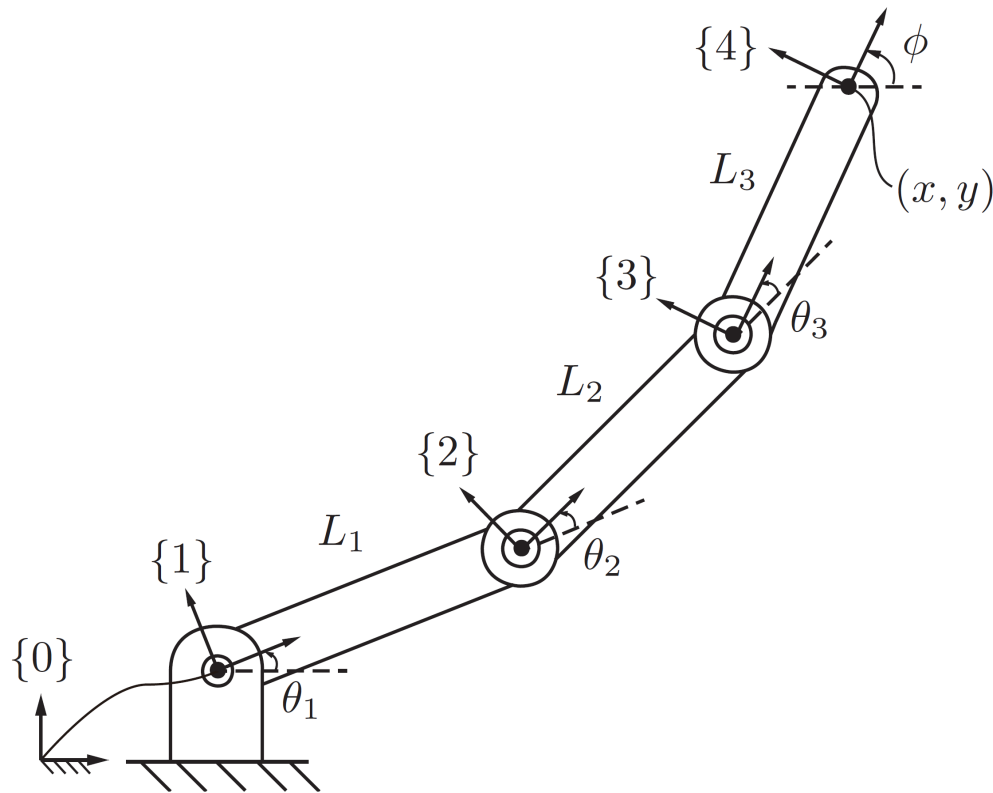
- Forward kinematics of a robot: calculation of the position and orientation of its end-effector from its joint coordinates  $\theta$
- Recall robot links and joints



# Forward Kinematics



# Forward Kinematics

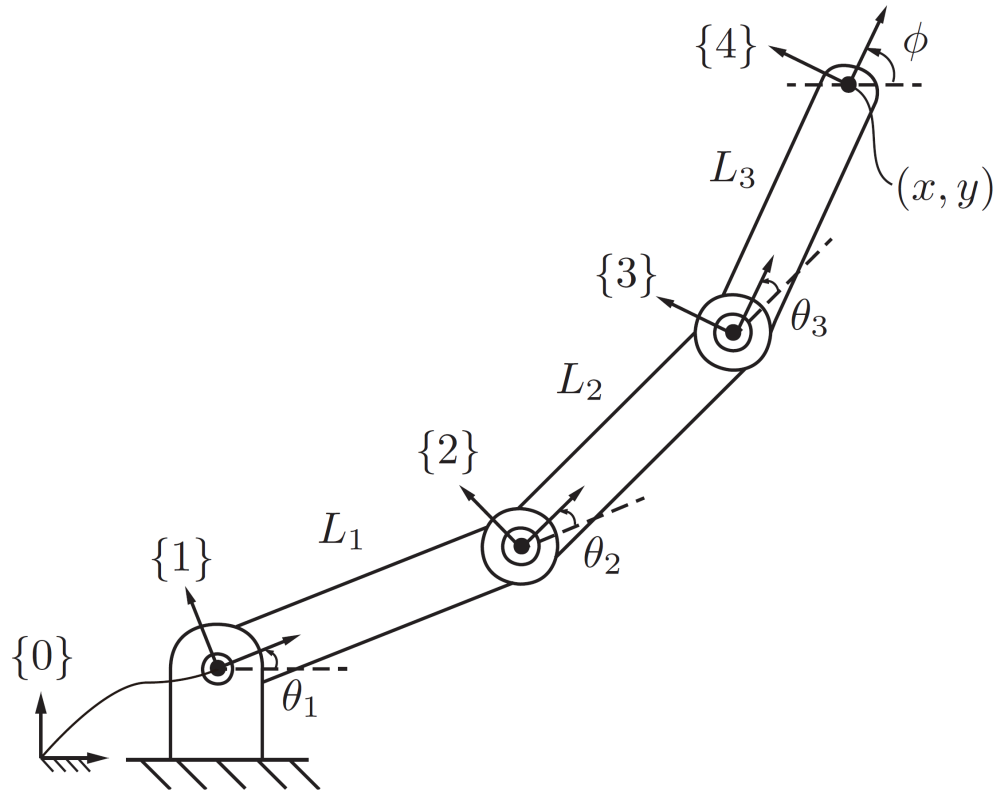


- End-effector frame {4}
- Joint angles  $(\theta_1, \theta_2, \theta_3)$
- Position and orientation of the end-effector frame in base frame {0}

$$\begin{aligned}x &= L_1 \cos \theta_1 + L_2 \cos(\theta_1 + \theta_2) + L_3 \cos(\theta_1 + \theta_2 + \theta_3), \\y &= L_1 \sin \theta_1 + L_2 \sin(\theta_1 + \theta_2) + L_3 \sin(\theta_1 + \theta_2 + \theta_3), \\ \phi &= \theta_1 + \theta_2 + \theta_3.\end{aligned}$$

Forward kinematics of a 3R planar open chain.

# Forward Kinematics



Forward kinematics of a 3R planar open chain.

- General cases
  - Attaching frames to links
  - Using homogeneous transformations

$$T_{04} = T_{01}T_{12}T_{23}T_{34}$$

$$T_{01} = \begin{bmatrix} \cos \theta_1 & -\sin \theta_1 & 0 & 0 \\ \sin \theta_1 & \cos \theta_1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad T_{12} = \begin{bmatrix} \cos \theta_2 & -\sin \theta_2 & 0 & L_1 \\ \sin \theta_2 & \cos \theta_2 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$T_{23} = \begin{bmatrix} \cos \theta_3 & -\sin \theta_3 & 0 & L_2 \\ \sin \theta_3 & \cos \theta_3 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad T_{34} = \begin{bmatrix} 1 & 0 & 0 & L_3 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$T_{i-1,i}$  Depends only on the joint variable  $\theta_i$

# Forward Kinematics

- Using homogeneous transformations
  - Need to define the coordinates of frames
- Denavit-Hartenberg Parameters

Northwestern Engineering's legacy in robotics started in the 1950s when Dick Hartenberg, a professor, and Jacques Denavit, a PhD student, developed a way to represent mathematically how mechanisms move <https://robotics.northwestern.edu/history.html>

# Summary

- Transformations
- Forward kinematics

# Further Reading

- Chapter 4 and Appendix C in Kevin M. Lynch and Frank C. Park. Modern Robotics: Mechanics, Planning, and Control. 1st Edition, 2017.
- J. Denavit and R. S. Hartenberg. A kinematic notation for lower-pair mechanisms based on matrices. ASME Journal of Applied Mechanics, 23:215-221, 1955.